Summary in Graph

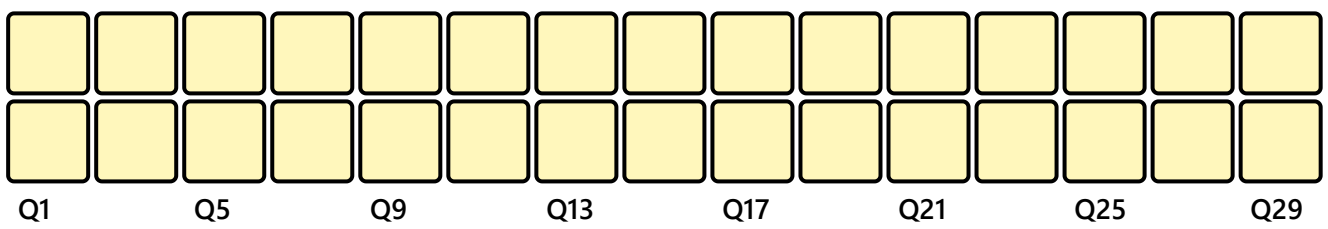
Exam Summary (GO Classes CS Test Series 2026 | Data Structures | Subject Wise Test 2)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	30 20 + 10
Total Marks:	40 20 + 20
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (48) out of 477 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose you have a stack in which the values 1 through 5 must be pushed on the stack in that order, but that an item on the stack can be popped at any time. Which of the following is valid sequence of push and pop operations such that the values are popped in the following order:

i. 23541
ii. 43521

iii. 21534

A. (i) and (ii)
B. (ii) and (iii)
C. (i) and (iii)
D. (i), (ii) and (iii)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following algorithm to check whether a string of parentheses is valid and balanced:

- Whenever an opening bracket ' (' is encountered, it is pushed onto the stack.
- Whenever a closing bracket ') ' is encountered, a bracket is popped from the stack.

What is the maximum number of characters that will appear on the stack at any time when the algorithm analyzes the string: (()()((()))(()))

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 03sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following pseudocode of a recursive function that takes a pointer to Node which is the head of a NULL-terminated singly linked list, and a key x.

```
Procedure(head, key):
    If (head == NULL) return NULL

    If (head.key == x)
        node *temp = head.next
        delete head
        return Procedure(temp, key)

    Else
        head.next = Procedure(head.next, key)
        return head;
```

Which of the following is True about the function?

A. The function deletes the first node of a singly linked list that matches the given key.
B. The function deletes every node of a singly linked list that matches the given key.
C. The function attempts to delete the first node of a singly linked list that matches but the implementation is wrong.
D. The function attempts to delete every node of a singly linked list that matches but the implementation is wrong.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

The following nonrecursive partial pseudocode reverses a list represented with a doubly linked list using a single pass through the list.

head: points to the head of doubly linked list.
prev: node's pointer to the previous node.
next: node's pointer to the next node.

```
Procedure(head):
    if (head == NULL or head.next == NULL) return

    previousNode = head
    currentNode = previousNode.next

    While ( _____(i)_____ ) {
        nextNode = currentNode.next

        currentNode.prev = ______(ii)______
        currentNode.next = previousNode

        previousNode = currentNode
        currentNode = nextNode
    }
```

Which of the following are appropriate choice for (i) and (ii) respectively?

A. (i) `currentNode != NULL`
B. (ii) `currentNode.next`
C. (i) `currentNode.next != NULL`
D. (ii) `nextNode`

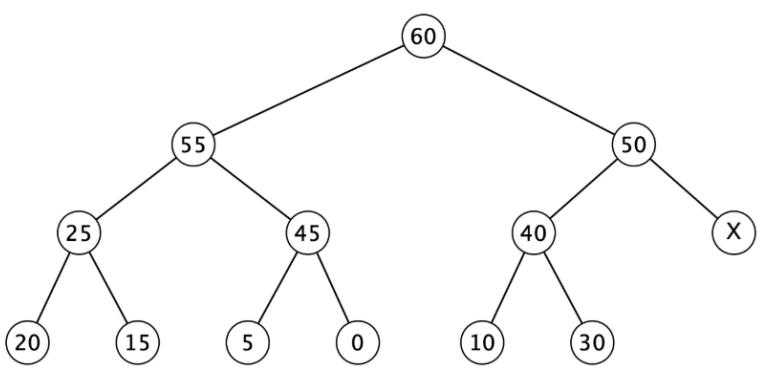
Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Suppose we have 4 distinct positive keys. All possible binary heaps are constructed by using these keys. let 'A' denotes the number of max-heap and ' B ' denotes the number of min-heap constructed, then $A+3B=$ _____

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures



If the last inserted key in Heap is 50 then what is the possible value(s) of X?

A. 35
B. 40
C. 45

D. 50

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

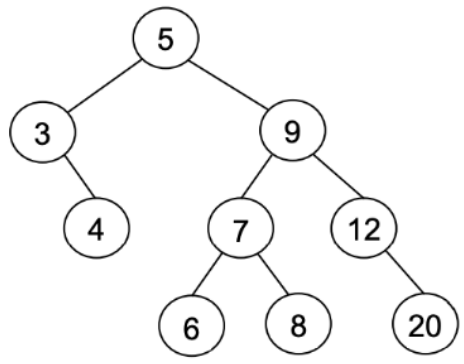
Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

The number of ways to create a binary heap tree with elements $21, 23, 53, 22, 14, 15, 90, 32$?

Your Answer: | Correct Answer: 210 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The binary search tree shown below was constructed by inserting a sequence of items into an empty tree



Which of the following input sequences will not produce this binary search tree?

A. 5
3
4
9
12
7
8
6
20

B. 5
9
3
7
6
8
4
12
20

C. 5
9
7
8
6
12
20
3
4

D. 5
9
3
6
7
8
4
12
20

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a binary search tree (BST) with n elements and height $h = O(\log(n))$, the correct asymptotic bound to find the $k\text{-}th(k < n)$ largest element in the tree is _____

A. $O(\mathrm{k})$
B. $O(\mathrm{h})$
C. $O(\mathrm{h}+\mathrm{k})$
D. $O(\mathrm{n})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider a Binary Search Tree (BST). Which of the following statements is/are TRUE regarding the inorder predecessor and inorder successor of a node being deleted?

A. Inorder predecessor of a deleting node is always either a leaf node or node without the right child.
B. Inorder successor of a deleting node is always either a leaf node or node without the right child.
C. Inorder predecessor of a deleting node is always either a leaf node or node without the left child.
D. Inorder successor of a deleting node is always either a leaf node or node without the left child.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A rooted tree is called a 3-ary tree if every node has either 0 or 3 children. Consider such a 3 -ary tree with n total nodes. If the number of leaf nodes in the tree is L, then the number of internal nodes in the tree is:

A. $\frac{L-1}{2}$

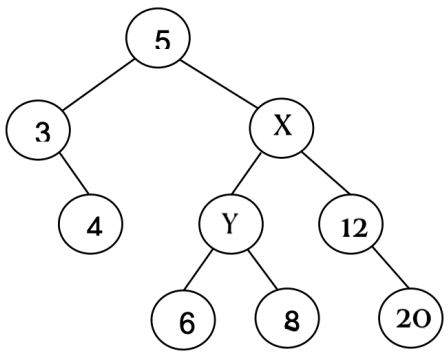
B. $\frac{L-1}{3}$

C. $\frac{3L-1}{2}$

D. $\frac{n-1}{L}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures



The graph shown above is a binary search tree of distinct positive integers. What will be the minimum possible difference of unknown nodes X to Y ?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose you combine open addressing with a limited form of chaining. You build an table with m slots that can store two keys in each slot.
1. Hash (key, probe number) to one of the m slots.
2. If the slot has less than two keys, insert it there.
3. Otherwise, increment the probe number and go to step 1.
Assume that there are exactly k slots in the table that are completely full. What is the probability that the first probe is successful?

A. $(\mathrm{m}-\mathrm{km})/\mathrm{d}$
B. k/m
C. $(\mathrm{m}-2\mathrm{k})/2\ \mathrm{m}$
D. $(\mathrm{m}-\mathrm{k})/\mathrm{m}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

A linear-probing hash table of length 10 uses the hash function $\mathbf{h}(\mathbf{k}) = \mathbf{k}$ mod 10 to hash integer keys between 0 and 100. After inserting 8 keys into an empty hash table, the table is as shown below.

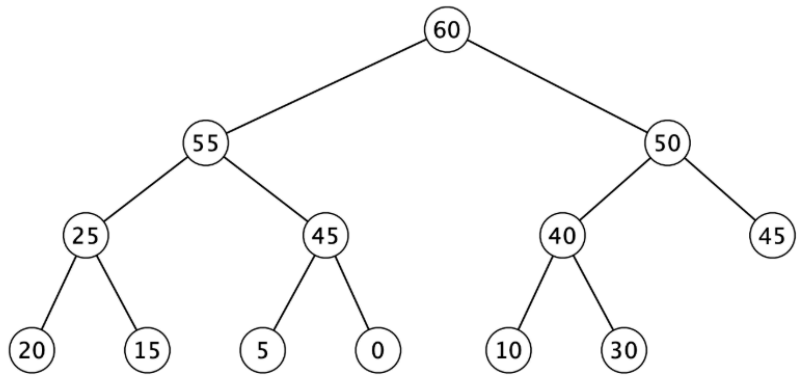
index	0	1	2	3	4	5	6	7	8	9
value	20	19	–	–	14	44	6	74	86	59

Which of the following choices gives a possible order in which the keys could have been inserted into an initially empty hash table? Assume that the initial length of the array is 10 and that it neither grows nor shrinks.

A. $14, 44, 6, 74, 86, 20, 59, 19$
B. $59, 6, 14, 20, 44, 19, 74, 86$
C. $20, 59, 6, 74, 14, 86, 44, 19$
D. $6, 59, 14, 20, 19, 44, 86, 74$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures



Suppose that the last operation in the binary heap was a call to insert(). Which key/s could have been the last one inserted?

A. 30
B. 40
C. 50
D. 60

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

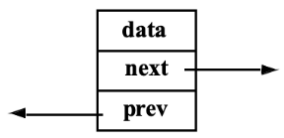
Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 2nd largest element of the tree is stored at Index .

A. 254
B. 255
C. 510
D. 511

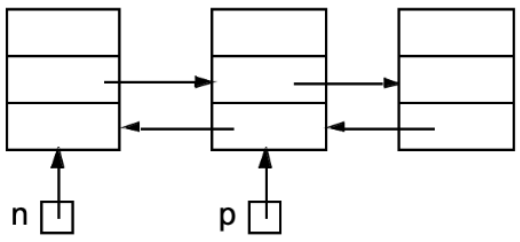
Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Nodes for a doubly linked list are defined to have the following structure:



The next instance variable stores a reference to the next node in the list, and the prev instance variable refers to the previous node in the list.
Below is a list of three of these nodes, along with two reference variables, n and p, that refer to specific nodes in the list.



Which of the following expressions does not refer to the third node in the list?

A. p.next
B. n.next.next
C. p.prev.next

D. n.next.next.prev.next

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Let L be a singly-linked list and X and Y be additional pointer variables such that X points to the first node of L and Y points to the last node of L. Which of the following operations cannot be done in time that is bounded above by a constant?

A. Delete the first node of L
B. Delete the last node of L
C. Swap first and last node of L
D. Swap the first two nodes of L

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following pseudocode:

```
procedure MYSTERY(root)
    if root.left is NULL then
        return root
    else
        return MYSTERY(root.left)
end procedure
```

Which of the following statements is/are **True** after the complete execution of the `MYSTERY` function called with the reference to the root of a Binary Search Tree of *n* nodes?

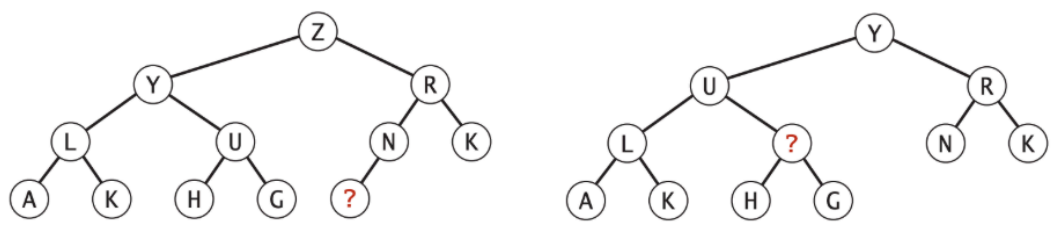
I. It returns the first node visited by an inorder traversal.
II. It returns the last node visited by a postorder traversal.
III. It returns the last node visited by a preorder traversal.

A. I
B. II
C. III
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

A delete operation in the max-heap of distinct keys results in a max-heap in right.



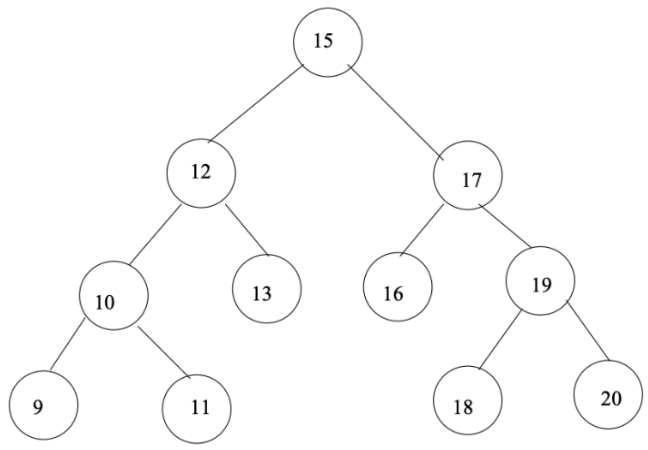
Which of the keys below could be the one labelled with a question mark?

A. I
B. J
C. L
D. M

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a Binary Search Tree(BST) containing integer keys. A Postorder traversal of the BST is to be performed, but only keys k satisfying $12 < k < 20$ are to be printed.



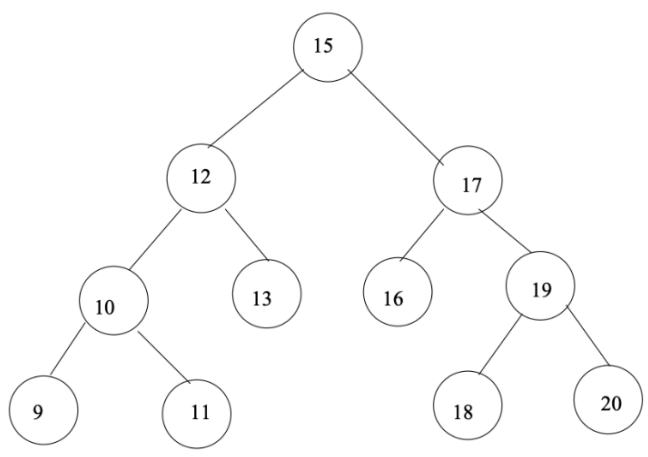
Which of the following sequences represents the **correct output** of this traversal?

A. $13, 15, 16, 17, 18, 19$
B. $15, 13, 17, 16, 19, 18$
C. $13, 16, 18, 19, 17, 15$
D. $15, 17, 13, 16, 19, 18$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Let T be a Binary Search Tree (BST) with distinct integer key:



The following operations(OPi) are considered independently (i.e. apply each operation to the original tree T), mark the TRUE Claim(s).

A. OP1: Delete key 15 from T. Claim: resulting BST is AVL-balanced.
B. OP2: Insert key 8 into T. Claim: resulting BST is AVL-balanced.
C. OP3: Insert key 14 in T. Claim: resulting BST is AVL-balanced.
D. OP4: Delete key 13 from T. Claim: resulting BST is AVL-balanced.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider an initially empty Binary Search Tree (BST). Insert the key values in the given order.

$$8, 10, 3, 6, 4, 7, 1, 14, 13$$

Determine the total number of key comparisons performed during the construction of the BST.

Your Answer: | Correct Answer: 17 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following statement(s) is TRUE about Binary Search Trees (BSTs):

A. In an AVL tree, during the insertion, at most two rotations are needed.
B. In a BST, we can find the next smallest element to a given element in $O(1)$ time.
C. Deleting a node and merging immediately results in the same shape of tree.
D. The height of any Binary Search tree with n nodes is $\Omega(logn)$.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider an array based implementation of a stack, and suppose that it is initially empty. Upon n push operations the array will be resized in such a way that the running time per push is $O(1)$ per operation, on average. How many times is the array resized over the n pushes, using this scheme?

A. $O(1)$

B. $O(\log n)$

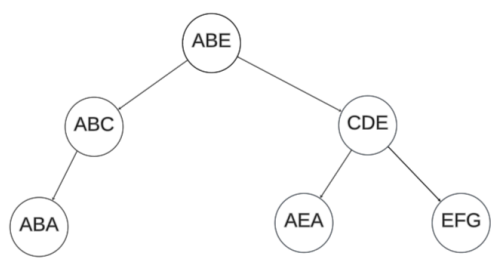
C. $O(\mathrm{n})$

D. $O\left(n^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a Blanched Binary Search Tree of **n** students' with keys as their names, stored in lexicographical ordering. Let k be the number of students whose name starts with prefix **X**.
For eg. a class having 6 students with names { ABE , CDE , AEA , ABC , EFG , ABA}, there are $\mathrm{k} = 3$ students whose name starts with prefix X=AB.



Which of the following is the correct time complexity to find all the students with prefix X.

A. $\theta(n \cdot \log \mathrm{k} + \mathrm{k})$
B. $\theta(n + \mathrm{k})$
C. $\theta(\log n + \mathrm{k})$
D. $\theta(\mathrm{k} \cdot \log \mathrm{n} + \mathrm{k})$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a hash table with n buckets, where linear probing is used to resolve collisions. The hash function is such that the probability that a key value is hashed to a particular bucket is $\frac{1}{n}$. The hash table is initially empty and distinct values are inserted in the table.
What is the probability that the given two distinct keys will be inserted at consecutive slots successfully?

A. $\frac{2}{n^2}$

B. $\frac{2}{n}$

C. $\frac{3}{n}$

D. $\frac{3}{n^2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a hash function that distributes keys uniformly. The hash table contains m slots, if chaining is used for collision resolution then the total number of available probe sequences for n keys?

A. $\mathrm{m}!$

B. m^2

C. m^n

D. m

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

A queue is implemented using two stacks, stack1 and stack2. The pseudocode for the implementation is given below:

```
enqueue(x){
    stack1.push(x);
}

dequeue(){
    if( stack2.empty() ){
        while( !stack1.empty() ){
            y = ____(1)____
            _________(2)____
        }
    }
    return stack2.pop();
}
```

Which of the following statements is/are correct?

A. The above two stack queue implementations is incorrect .
B. (1) stack1.pop(); (2) stack2.push(y);
C. (1) stack2.pop(); (2) stack1.push(y);
D. The running time complexity of dequeue operation over a long sequence is $\mathrm{O}(\mathrm{n})$.

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

You are given a traversal sequence S of distinct positive integers and an unknown tree T.
Which of the following statements is/are TRUE?

A. If T is a binary search tree, then inorder traversal of T can be determined.
B. If T is a complete binary tree and S is preorder traversal, then inorder traversal of T can be determined.

C. If T is a complete binary tree and S is inorder traversal, then preorder traversal of T can be determined.

D. If T is a binary search tree and S is preorder traversal, then postorder traversal of T can be determined.

Your Answer: | **Correct Answer: A;B;C;D** | Not Attempted | Time taken: 00min 00sec | Discuss

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